

SMART INDIA HACKATHON 2026



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HACKATHON
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- **Problem Statement ID** – SIH 26213
- **Problem Statement Title-** AI Powered Smart
Badminton Fitness and Performance Analytics System
- **Theme-** Fitness & Sports
- **PS Category-** Hardware
- **Team Name :** NOVA TITANS



AI Powered Smart Badminton Fitness and Performance Analytics System



Problem / Challenge :

1. Players evaluated mostly via coach observation without objective, measurable metrics
2. Missing personalized progress analytics to track player improvement across sessions.
3. Lack of data on stroke speed, intensity, consistency, and fatigue during training.

Existing Solutions :

1. Current systems track basic stroke counts, swing speed, and activity using standard wearable sensors and IMUs.

Gap Analysis

1. Personalized Player Assessment
2. Integrated Skill + Fitness Analysis
3. Compact & Affordable Racket System

Our Proposed Solution :

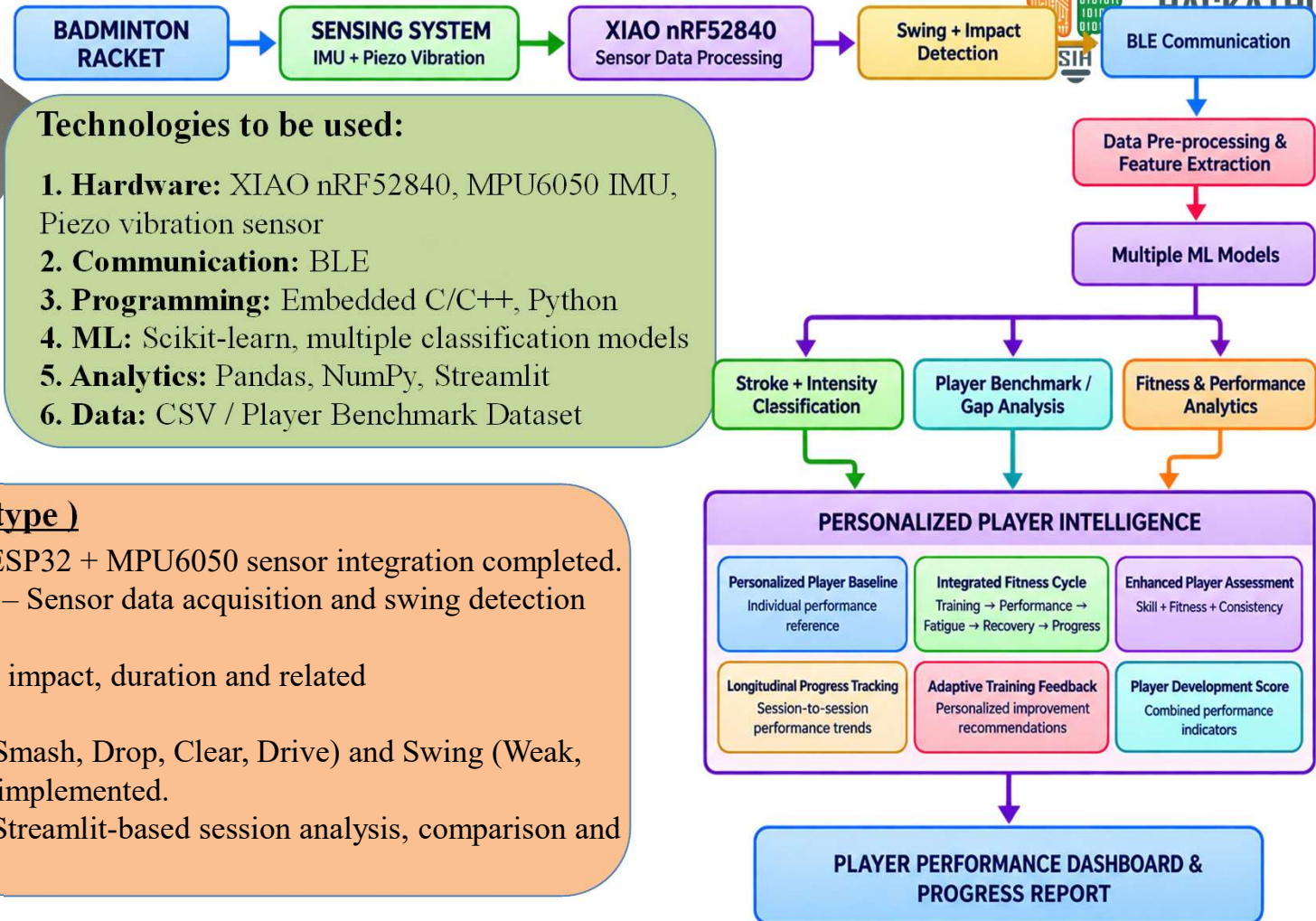
1. **AI/ML-Based Player Assessment** – Analyse strokes and performance using sensor data.
2. **Personalized Training Analytics** – Track individual baselines, strengths and improvement.
3. **Fitness & Performance Monitoring** – Monitor intensity, consistency, workload and fatigue.
4. **Smart Player Profiling** – Combine session and benchmark data for player insights.
5. **Affordable Smart-Racket Platform** – Provide a compact solution for training and assessment.

Innovation and Uniqueness :

1. **Dual-Signal Stroke Validation** – IMU + piezo confirms racket–shuttle impact.
2. **Two-Stage Data Processing** – Threshold detection followed by ML analysis.
3. **Multi-Model Intelligence** – Separate ML models for stroke and intensity classification.
4. **Benchmark Gap Analysis** – Compare player data with trained pro-player reference data.
5. **V1 → V2 Evolution** – Upgrade the proven prototype with better sensing and connectivity.

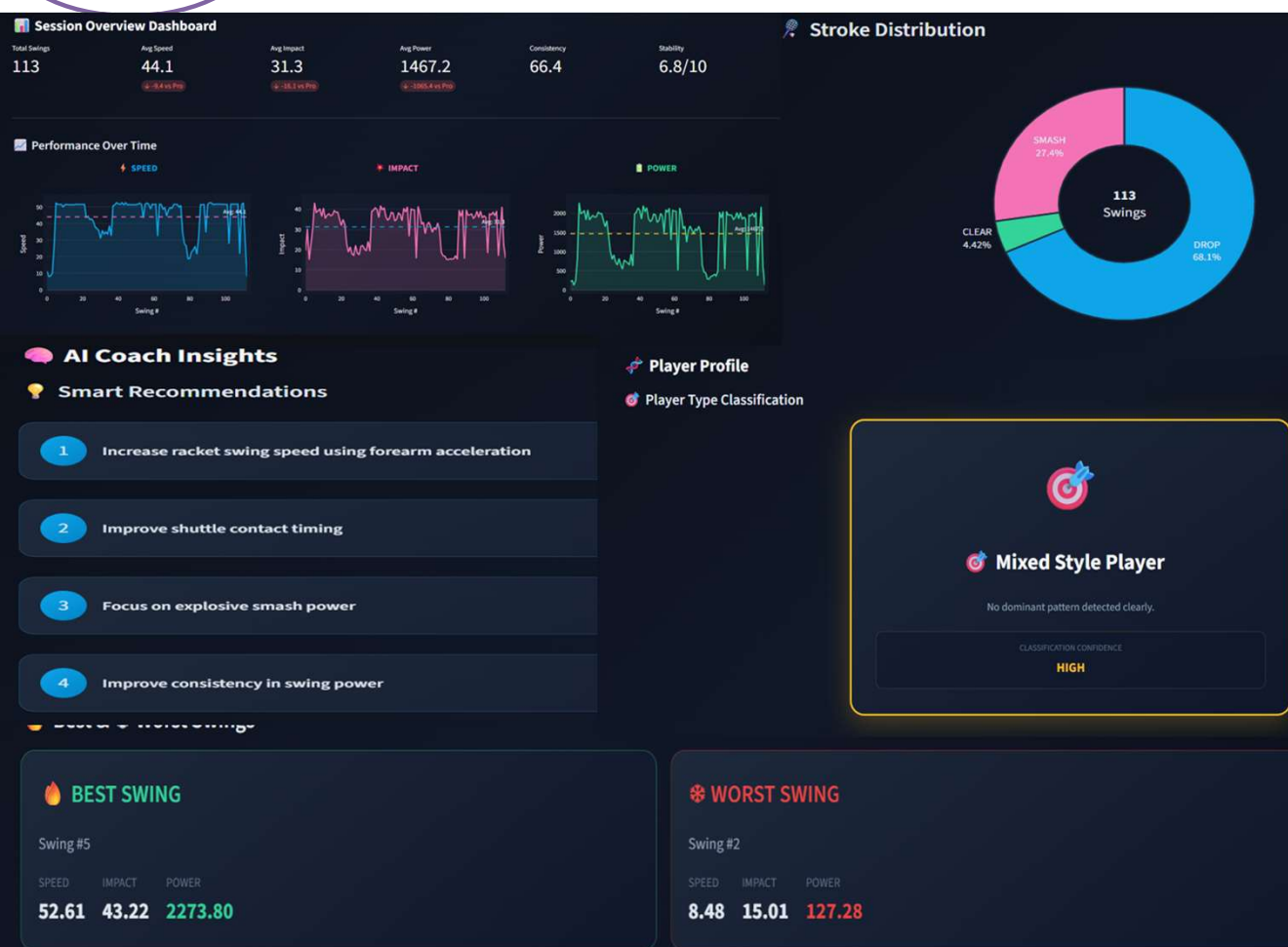
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TECHNICAL APPROACH



Version 1 (Working Prototype)

- Smart Racket Prototype** – ESP32 + MPU6050 sensor integration completed.
- Real-Time Swing Detection** – Sensor data acquisition and swing detection implemented.
- Feature Extraction** – Speed, impact, duration and related swing features calculated.
- ML Classification** – Stroke(Smash, Drop, Clear, Drive) and Swing (Weak, Medium, Strong) classification implemented.
- Performance Dashboard** – Streamlit-based session analysis, comparison and player insights developed.



Technical Feasibility

1. V1 prototype validates sensing, swing detection and ML processing.
2. V2 uses readily available IMU, Piezo and BLE hardware.
3. Modular design supports future sensor and analytics upgrades.

Challenges

1. **Sensor Noise:** Racket vibration may cause false impact detection.
2. **BLE & Power:** Data loss and battery consumption during continuous operation.
3. **Personalization:** Different playing styles make universal baselines difficult.

Solutions

1. **IMU + Piezo fusion,** filtering and player-specific calibration.
2. **Data buffering + SD/Wi-Fi fallback** for communication reliability.
3. **Adaptive power modes + multi-session baselines** for efficiency and personalization.

IMPACT AND BENEFITS

Target Audience Impact Analysis

Players & Beginners

Get measurable feedback on stroke quality, consistency and progress. Identify individual strengths and areas for improvement.

Coaches & Badminton Academies

Enable data-driven player evaluation beyond visual observation. Monitor player performance and development across training sessions. Identify training needs and support personalized coaching strategies.

Sports & Biomedical

Support early identification of performance decline and injury-risk indicators through changes in movement patterns.

Benefits Analysis

Training & Performance Benefits

Enables session-wise and long-term performance tracking. Helps establish measurable benchmarks for player development.

Biomedical & Sports Science Benefits

Generates structured datasets for sports biomechanics and movement research. Can support future integration with physiological sensors for more comprehensive fitness assessment

Economic Benefits

Affordable and portable compared with high-end sports analytics systems. Can be deployed across multiple players and training centres at relatively low hardware cost.

1. Wearable Sensing & AI

Jin, G., LI, X. Wearable sensing for badminton stroke recognition with one-dimensional convolutional neural network. *Sci Rep* **15**, 41236 (2025). <https://doi.org/10.1038/s41598-025-25158-2>

2. Badminton Biomechanics & Performance Dataset

Seong, M., Kim, G., Yeo, D. *et al.* MultiSenseBadminton: Wearable Sensor–Based Biomechanical Dataset for Evaluation of Badminton Performance. *Sci Data* **11**, 343 (2024). <https://doi.org/10.1038/s41597-024-03144-z>

3. AI-Based Badminton Motion Analysis

Zhu, X., Liu, L., Huang, J. *et al.* The analysis of motion recognition model for badminton player movements using machine learning. *Sci Rep* **15**, 19030 (2025). <https://doi.org/10.1038/s41598-025-02771-9>

Jacob, Alvin, Wan Nurshazwani Wan Zakaria, and Mohd Razali Bin Md Tomari. "Implementation of IMU sensor for elbow movement measurement of badminton players." *2016 2nd IEEE International Symposium on Robotics and Manufacturing Automation (ROMA)*. IEEE, 2016.

Li, Li, and Feng Li. "Recognizing badminton hitting movements Based on Multi-sensor trajectory recognition algorithm." *2023 3rd International Conference on Information Technology and Contemporary Sports (TCS)*. IEEE, 2023.